Raay: A Native-Language Voice Operating System for Windows

Repository: https://github.com/vakrahul/Raay | Built by: Rahul Vakiti and Samyutha

Why We Built Raay

Most people in India do not think in English prompts. When you put a laptop in front of a local kirana shopkeeper, our parents, or someone who speaks Telugu, Hindi, or Tamil at home, the biggest wall between them and the computer is the English keyboard. Even when people try modern AI chatbots like ChatGPT or Gemini, two things break immediately. First, chatbots are trapped inside a browser tab; if you ask a chatbot to open your local Excel grocery sheet and check today's UPI payments, or open MS Paint and draw a diagram, it simply says it cannot access your computer. Second, standard voice tools fail on how we actually talk in daily life, where we naturally mix Telugu, Hindi, or Tamil with English words in the same sentence, such as asking, "Rahul, grocery Excel file open cheyyi, indulo UPI payments enni?" or "Circle lopa unna red color ni yellow ki change cheyyi." We built Raay so that you do not have to type at all. You speak naturally to your Windows laptop in your own language, and the computer opens the applications and completes the work right in front of you.

How Raay Works with Gnani.ai and Pluggable Planners

Raay runs as a lightweight, always-on background voice service on Windows 11 powered by Gnani.ai's speech stack. A background microphone listener captures 16 kHz mono audio and streams it over WebSocket directly to Gnani Prisma STT (wss://api.vachana.ai/stt/v3/stream). By tuning the voice activity detection silence window to 280 milliseconds and enabling telephonic noise cancellation (nc=true), Prisma transcribes code-mixed Telugu, Hindi, Tamil, and English speech almost the instant you finish speaking, even in a noisy room. Before executing open-ended automation tasks, Raay passes the transcribed request through a pluggable model adapter layer (src/providers/). Developers are not locked into a single model: you can plug in Anthropic Claude, OpenAI GPT, Google Gemini, or a local Ollama model to break complex requests into step-by-step plans before desktop execution begins, while routine spreadsheet math and canvas edits take a direct deterministic path in under 10 milliseconds. Once the desktop action completes, Gnani Timbre v3.0 TTS (POST /api/v1/tts/sse) speaks the result back in the exact same language using natural Indian prosody and rupee/lakh/crore cadence, backed by a local SHA-256 audio cache and automatic microphone muting during speaker playback.

Real Desktop Execution on Windows 11

Instead of relying on fragile screen-coordinate macros that break whenever a window moves, Raay hooks directly into the Windows UI Automation (IUIAutomation) accessibility tree, Win32 window handles (HWND), and Microsoft Excel COM automation. In our live tests across five everyday desktop workflows where standard chatbots score zero out of five, Raay completed all five end-to-end. When asked in Telugu to open the grocery bill and check UPI payments, Raay launches Microsoft Excel on a 25-item kirana billing sheet, renders live bar and pie charts on screen, and replies in Telugu that 12 UPI items total Rs. 2,478 out of a Rs. 4,741 grand total with Rs. 249 in GST. When asked to open MS Paint and draw a red circle with a heart beside it, Raay inspects the Paint ribbon via UIA, selects the ellipse and heart tools, picks the red colour swatch, and draws both objects on the canvas. When interrupted mid-workflow with "Change the color of the circle to yellow" or "Aapu! Undo chey," Raay halts within 200 milliseconds, triggers Ctrl+Z undo, selects the yellow swatch, and redraws the circle in place. It also exports active Paint designs into HTML5, SVG, and JSON project files opened directly in Visual Studio Code, opens websites in Chrome by voice, and scans the local Downloads folder to summarize the latest report into Notepad. All API keys are encrypted locally using the Windows Data Protection API (DPAPI) so no credentials ever sit in plaintext files.

Who Built What

Rahul Vakiti designed the system architecture and built the Gnani Prisma WebSocket STT and Timbre v3.0 SSE TTS streaming pipeline, the always-on background microphone listener, the Windows 11 UIA, Win32, and Excel COM execution controllers, the sub-200ms voice interrupt and undo engine, and the Windows DPAPI credential vault. Samyutha built the multilingual conversational planner across Telugu (te-IN), Hindi (hi-IN), Tamil (ta-IN), Hinglish (hi-en), and Indian English (en-IN), the Indian script verification rules, the 25-item kirana and grocery GST analytics model, and the live Web UI interface.